# VISVESVARAYA TECHNOLOGICAL UNIVERSITY

**"JnanaSangama", Belgaum- 590014, Karnataka.**

**LAB RECORD**

**on**

# Big Data Analytics (23CS6PCBDA)

***Submitted by***

**Sareddy Poojya Sree (1BM23CS303)**

***in partial fulfillment for the award of the degree of***

**BACHELOR OF ENGINEERING**

***in***

**COMPUTER SCIENCE AND ENGINEERING**

**B.M.S. COLLEGE OF ENGINEERING**
**(Autonomous Institution under VTU)**
**BENGALURU - 560019**
**January 2026 – May 2026**

# B.M.S. College of Engineering

**Bull Temple Road, Bangalore 560019**
(Affiliated to Visvesvaraya Technological University, Belgaum)

## Department of Computer Science and Engineering

## CERTIFICATE

This is to certify that the Lab work entitled "Big Data Analytics" carried out by **Sareddy Poojya Sree (1BM23CS303),** who is bonafide student of **B.M.S. College of Engineering.** It is in partial fulfilment for the award of **Bachelor of Engineering in Computer Science and Engineering** of the Visvesvaraya Technological University, Belgaum during the year 2026. The Lab report has been approved as it satisfies the academic requirements in respect of a Big Data Analytics (23CS6PCBDA) work prescribed for the said degree.

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Github Link: https://github.com/poojya100/1BM23CS303_BDA_LAB

**Course Outcomes (COs):**

| | |
|---|---|
| **CO1** | Apply the concept of NoSQL, Hadoop or Spark for a given task |
| **CO2** | Analyse big data analytics mechanisms that can be applied to obtain solution for a given problem. |
| **CO3** | Design and implement solutions using data analytics mechanisms for a given problem. |

# LABORATORY PROGRAM – 1

## MongoDB- CRUD Operations Demonstration (Practice and Self Study)

COMMAND WITH OUTPUT - USING ATLAS

```
>_MONGOSH
> db["Employee_Info"].find()
<
> use Employee
< switched to db Employee
> db.createCollection("Employee_Info")
< { ok: 1 }
> db.Employee_Info.createIndex({ Emp_Id: 1 }, { unique: true })
✖ > MongoServerError[DuplicateKey]: Index build failed: 5778370
> db.Employee_Info.insertMany([
    {
      _id: 121,
      Emp_Name: "Alice",
      Designation: "Software Engineer",
      Date_of_Joining: ISODate("2021-06-15"),
      Salary: 75000,
      Dept_Name: "IT"
    },
    {
      _id: 122,
      Emp_Name: "Bob",
      Designation: "Senior Engineer",
      Date_of_Joining: ISODate("2020-03-10"),
      Salary: 90000,
      Dept_Name: "R&D"
    },
    {
      _id: 123,
      Emp_Name: "Charlie",
      Designation: "Manager",
      Date_of_Joining: ISODate("2019-01-20"),
      Salary: 105000,
      Dept_Name: "HR"
    }
  ])
```

```
< {
    acknowledged: true,
    insertedIds: {
      '0': 121,
      '1': 122,
      '2': 123
    }
  }
> db.Employee_Info.updateOne(
    { _id: 121 },
    {
      $set: {
        Emp_Name: "Alice Johnson",
        Dept_Name: "Engineering"
      }
    }
  )
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 1,
    modifiedCount: 1,
    upsertedCount: 0
  }
> db.Employee_Info.find().sort({ Salary: 1 })
< {
    _id: 121,
    Emp_Name: 'Alice Johnson',
    Designation: 'Software Engineer',
    Date_of_Joining: 2021-06-15T00:00:00.000Z,
    Salary: 75000,
    Dept_Name: 'Engineering'
  }
  {
    _id: 122,
    Emp_Name: 'Bob',
    Designation: 'Senior Engineer',
```

```
  {
    _id: 122,
    Emp_Name: 'Bob',
    Designation: 'Senior Engineer',
    Date_of_Joining: 2020-03-10T00:00:00.000Z,
    Salary: 90000,
    Dept_Name: 'R&D'
  }
  {
    _id: 123,
    Emp_Name: 'Charlie',
    Designation: 'Manager',
    Date_of_Joining: 2019-01-20T00:00:00.000Z,
    Salary: 105000,
    Dept_Name: 'HR'
  }
> db.Employee_Info.find().sort({ Salary: -1 })
< {
    _id: 123,
    Emp_Name: 'Charlie',
    Designation: 'Manager',
    Date_of_Joining: 2019-01-20T00:00:00.000Z,
    Salary: 105000,
    Dept_Name: 'HR'
  }
  {
    _id: 122,
    Emp_Name: 'Bob',
    Designation: 'Senior Engineer',
    Date_of_Joining: 2020-03-10T00:00:00.000Z,
    Salary: 90000,
    Dept_Name: 'R&D'
  }
  {
    _id: 121,
    Emp_Name: 'Alice Johnson',
    Designation: 'Software Engineer',
```

```
{
  _id: 121,
  Emp_Name: 'Alice Johnson',
  Designation: 'Software Engineer',
  Date_of_Joining: 2021-06-15T00:00:00.000Z,
  Salary: 75000,
  Dept_Name: 'Engineering'
}
> db.Employee_Info.updateOne(
  { _id: 121 },
  {
    $addToSet: {
      Projects: { $each: ["Payroll System", "Cloud Migration"] }
    }
  }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
> db.Employee_Info.updateMany(
  {},
  {
    $set: {
      expireAt: new Date(Date.now() + 15000)
    }
  }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 3,
  modifiedCount: 3,
  upsertedCount: 0
}
```

```
> db.Employee_Info.updateMany(
    {},
    {
      $set: {
        expireAt: new Date(Date.now() + 15000)
      }
    }
  )
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 3,
    modifiedCount: 3,
    upsertedCount: 0
  }
> db.Employee_Info.updateMany(
    {},
    {
      $set: {
        expireAt: new Date(Date.now() + 15000)
      }
    }
  )
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 3,
    modifiedCount: 3,
    upsertedCount: 0
  }
> db.Employee_Info.createIndex(
    { expireAt: 1 },
    { expireAfterSeconds: 0 }
  )
< expireAt_1
>
Employee>
```

# MongoDB- CRUD Operations Demonstration (Practice and Self Study)

COMMAND WITH OUTPUT - USING UBUNTU TERMINAL

```
MyDataBase> use MyDataBase
already on db MyDataBase
MyDataBase> show collections
Customers
NewStudent
Student
MyDataBase> db.Student.find();
[
  {
    _id: 1,
    studName: 'Michellejacintha',
    Grade: 'VII',
    Hobbies: 'InternetSurfing'
  },
  { _id: 3, Grade: 'VII', StudName: 'AryanDavid', Hobbies: 'Skating' },
  { _id: 2, Grade: 'VIII', StudName: 'Ram', Hobbies: 'Learning' }
]
```

```
test> use MyDataBase
switched to db MyDataBase
MyDataBase> show collections
NewStudent
NewStudent2
Student
MyDataBase> db.NewStudent2.drop();
true
MyDataBase> db.createCollection("Customers");
{ ok: 1 }
MyDataBase> db.Customers.insertMany([{cust_id:1,Balance:200, Type:"S"},]);
{
  acknowledged: true,
  insertedIds: { '0': ObjectId('67d00571207666297fa3b81a') }
}
MyDataBase> db.Customers.insert({cust_id:1,Balance:1000, Type:"Z"})
DeprecationWarning: Collection.insert() is deprecated. Use insertOne, insertMany, or bulkWrite.
{
  acknowledged: true,
  insertedIds: { '0': ObjectId('67d0058f207666297fa3b81b') }
}
MyDataBase> db.Customers.insert({cust_id:2,Balance:100, Type:"Z"});
{
  acknowledged: true,
  insertedIds: { '0': ObjectId('67d0059c207666297fa3b81c') }
}
MyDataBase> db.Customers.insert({cust_id:2,Balance:1000, Type:"C"});
{
  acknowledged: true,
  insertedIds: { '0': ObjectId('67d005a5207666297fa3b81d') }
}
MyDataBase> db.Customers.insert({cust_id:2,Balance:500, Type:"C"});
{
  acknowledged: true,
  insertedIds: { '0': ObjectId('67d005ad207666297fa3b81e') }
}
MyDataBase> db.Customers.insert({cust_id:2,Balance:50, Type:"S"});
{
  acknowledged: true,
  insertedIds: { '0': ObjectId('67d005b2207666297fa3b81f') }
}
MyDataBase> db.Customers.insert({cust_id:3,Balance:500, Type:"Z"});
{
  acknowledged: true,
  insertedIds: { '0': ObjectId('67d005ba207666297fa3b820') }
}
```

```
MyDataBase> db.Customers.aggregate([
...   {
...     $group: {
...       _id: "$cust_id",              // Group by cust_id
...       minAccBal: { $min: "$Balance" }, // Find the minimum Balance
...       maxAccBal: { $max: "$Balance" }  // Find the maximum Balance
...     }
...   }
... ]);
[
  { _id: 3, minAccBal: 500, maxAccBal: 500 },
  { _id: 2, minAccBal: 50, maxAccBal: 1000 },
  { _id: 1, minAccBal: 200, maxAccBal: 1000 }
]
```

```
MyDataBase> db.Customers.aggregate([
...   { $match: { Type: "Z" } },
...   { $group: { _id: "$cust_id", TotAccBal: { $sum: "$Balance" } } },
...   { $match: { TotAccBal: { $gt: 1200 } } }
... ]);
```

```
MyDataBase> db.Customers.aggregate([
...   { $match: { Type: "Z" } },
...   {
...     $group: {
...       _id: "$cust_id",
...       TotAccBal: { $sum: "$Balance" }
...     }
...   },
...   {
...     $match: {
...       TotAccBal: { $gt: 200 }
...     }
...   }
... ]);
[ { _id: 3, TotAccBal: 500 }, { _id: 1, TotAccBal: 1000 } ]
```

```
bmscecse@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ mongoimport --host localhost --db MyDataBase --coll
ection NewStudent2 --type=csv --file /home/bmscecse/Desktop/135.txt --headerline
2025-03-11T14:55:05.192+0530    connected to: mongodb://localhost/
2025-03-11T14:55:05.360+0530    3 document(s) imported successfully. 0 document(s) failed to import.
bmscecse@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ mongoexport --host localhost --db MyDataBase --coll
ection NewStudent2 --type=json --file /home/bmscecse/Desktop/135.txt
2025-03-11T14:55:24.438+0530    error parsing command line options: unknown option "file"
2025-03-11T14:55:24.438+0530    try 'mongoexport --help' for more information
bmscecse@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ mongoexport --host localhost --db MyDataBase --coll
ection NewStudent2 --type=json --out /home/bmscecse/Desktop/135.txt
2025-03-11T14:55:32.771+0530    connected to: mongodb://localhost/
2025-03-11T14:55:32.780+0530    exported 3 records
bmscecse@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$
```

# LABORATORY PROGRAM – 2

## Perform the following DB operations using Cassandra

## Questions:

a) Create a keyspace by name Employee
b) Create a column family by name

- Employee-Info with attributes
- Emp_Id Primary Key, Emp_Name,
- Designation, Date_of_Joining,
- Salary, Dept_Name

c) Insert the values into the table in batch
d) Update Employee name and Department of Emp-Id 121
e) Sort the details of Employee records based on salary
f) Alter the schema of the table Employee_Info to add a column Projects which stores a set of Projects done by the corresponding Employee.
g) Update the altered table to add project names.
h) Create a TTL of 15 seconds to display the values of Employees.

COMMAND WITH OUTPUT

```
mongosh mongodb://127.0.0.
icrosoft Windows [Version 10.0.26100.7623]
(c) Microsoft Corporation. All rights reserved.

:\Users\student>mongosh
urrent Mongosh Log ID: 6989a2879a3a13ba051e2620
onnecting to:          mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.5.10
sing MongoDB:          8.0.11
sing Mongosh:          2.5.10
ongosh 2.6.0 is available for download: https://www.mongodb.com/try/download/shell

or mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

-----
  The server generated these startup warnings when booting
  2026-02-09T08:35:42.510+05:30: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
-----

est> use Employee
witched to db Employee
mployee> db.Employee_Info.insertMany([
.. {
..   Emp_Id:121,
..   Emp_Name:"Amit",
..   Designation:"Software Engineer",
..   Date_of_Joining:new Date("2019-06-15"),
..   Salary:60000,
..   Dept_Name:"IT"
.. },
.. {
..   Emp_Id:122,
..   Emp_Name:"Ravi",
..   Designation:"Senior Engineer",
..   Date_of_Joining:new Date("2018-03-20"),
..   Salary:75000,
..   Dept_Name:"R&D"
.. },
.. {
..   Emp_Id:123,
..   Emp_Name:"Meera",
..   Designation:"Manager",
..   Date_of_Joining:new Date("2016-01-10"),
..   Salary:90000,
..   Dept_Name:"HR"
.. }
.. ])
..

 acknowledged: true,
 insertedIds: {
   '0': ObjectId('6989a2be9a3a13ba051e2621'),
   '1': ObjectId('6989a2be9a3a13ba051e2622'),
```

```
mongosh mongodb://127.0.0.
Employee> db.Employee_Info.find()
[
  {
    _id: ObjectId('6989a2be9a3a13ba051e2621'),
    Emp_Id: 121,
    Emp_Name: 'Amit',
    Designation: 'Software Engineer',
    Date_of_Joining: ISODate('2019-06-15T00:00:00.000Z'),
    Salary: 60000,
    Dept_Name: 'IT'
  },
  {
    _id: ObjectId('6989a2be9a3a13ba051e2622'),
    Emp_Id: 122,
    Emp_Name: 'Ravi',
    Designation: 'Senior Engineer',
    Date_of_Joining: ISODate('2018-03-20T00:00:00.000Z'),
    Salary: 75000,
    Dept_Name: 'R&D'
  },
  {
    _id: ObjectId('6989a2be9a3a13ba051e2623'),
    Emp_Id: 123,
    Emp_Name: 'Meera',
    Designation: 'Manager',
    Date_of_Joining: ISODate('2016-01-10T00:00:00.000Z'),
    Salary: 90000,
    Dept_Name: 'HR'
  }
]
Employee> db.Employee_Info.find().pretty()
[
  {
    _id: ObjectId('6989a2be9a3a13ba051e2621'),
    Emp_Id: 121,
    Emp_Name: 'Amit',
    Designation: 'Software Engineer',
    Date_of_Joining: ISODate('2019-06-15T00:00:00.000Z'),
    Salary: 60000,
    Dept_Name: 'IT'
  },
  {
    _id: ObjectId('6989a2be9a3a13ba051e2622'),
    Emp_Id: 122,
    Emp_Name: 'Ravi',
    Designation: 'Senior Engineer',
    Date_of_Joining: ISODate('2018-03-20T00:00:00.000Z'),
    Salary: 75000,
    Dept_Name: 'R&D'
  },
  {
```

```
mongosh mongodb://127.0.0.
Employee> db.Employee_Info.find().pretty()
[
  {
    _id: ObjectId('6989a2be9a3a13ba051e2621'),
    Emp_Id: 121,
    Emp_Name: 'Amit',
    Designation: 'Software Engineer',
    Date_of_Joining: ISODate('2019-06-15T00:00:00.000Z'),
    Salary: 60000,
    Dept_Name: 'IT'
  },
  {
    _id: ObjectId('6989a2be9a3a13ba051e2622'),
    Emp_Id: 122,
    Emp_Name: 'Ravi',
    Designation: 'Senior Engineer',
    Date_of_Joining: ISODate('2018-03-20T00:00:00.000Z'),
    Salary: 75000,
    Dept_Name: 'R&D'
  },
  {
    _id: ObjectId('6989a2be9a3a13ba051e2623'),
    Emp_Id: 123,
    Emp_Name: 'Meera',
    Designation: 'Manager',
    Date_of_Joining: ISODate('2016-01-10T00:00:00.000Z'),
    Salary: 90000,
    Dept_Name: 'HR'
  }
]
Employee> db.Employee_Info.updateOne(
...   {Emp_Id:121},
...   {$set:{Emp_Name:"Amit Kumar", Dept_Name:"Data Science"}}
... )
...
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
Employee> db.Employee_Info.find({Emp_Id:121})
[
  {
    _id: ObjectId('6989a2be9a3a13ba051e2621'),
    Emp_Id: 121,
    Emp_Name: 'Amit Kumar',
    Designation: 'Software Engineer',
    Date_of_Joining: ISODate('2019-06-15T00:00:00.000Z'),
    Salary: 60000,
```

```
mongosh mongodb://127.0.0.
Employee> db.Employee_Info.updateOne(
..   {Emp_Id:121},
..   {$set:{Emp_Name:"Amit Kumar", Dept_Name:"Data Science"}}
.. )
..
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
Employee> db.Employee_Info.find({Emp_Id:121})
[
  {
    _id: ObjectId('6989a2be9a3a13ba051c2621'),
    Emp_Id: 121,
    Emp_Name: 'Amit Kumar',
    Designation: 'Software Engineer',
    Date_of_Joining: ISODate('2019-06-15T00:00:00.000Z'),
    Salary: 60000,
    Dept_Name: 'Data Science'
  }
]
Employee> db.Employee_Info.find().sort({Salary:-1})
[
  {
    _id: ObjectId('6989a2be9a3a13ba051e2623'),
    Emp_Id: 123,
    Emp_Name: 'Meera',
    Designation: 'Manager',
    Date_of_Joining: ISODate('2016-01-10T00:00:00.000Z'),
    Salary: 90000,
    Dept_Name: 'HR'
  },
  {
    _id: ObjectId('6989a2be9a3a13ba051e2622'),
    Emp_Id: 122,
    Emp_Name: 'Ravi',
    Designation: 'Senior Engineer',
    Date_of_Joining: ISODate('2018-03-20T00:00:00.000Z'),
    Salary: 75000,
    Dept_Name: 'R&D'
  },
  {
    _id: ObjectId('6989a2be9a3a13ba051e2621'),
    Emp_Id: 121,
    Emp_Name: 'Amit Kumar',
    Designation: 'Software Engineer',
    Date_of_Joining: ISODate('2019-06-15T00:00:00.000Z'),
    Salary: 60000,
```

```
mongosh mongodb://127.0.0.
mployee> db.Employee_Info.updateOne(
..   {Emp_Id:121},
..   {$set:{
..     Projects:["AI-Analytics","Cloud-Migration","BigData-Pipeline"]
..   }}
.. )
..

 acknowledged: true,
 insertedId: null,
 matchedCount: 1,
 modifiedCount: 1,
 upsertedCount: 0

mployee> db.Employee_Info.find().sort({Salary:1})

 {
   _id: ObjectId('6989a2be9a3a13ba051e2621'),
   Emp_Id: 121,
   Emp_Name: 'Amit Kumar',
   Designation: 'Software Engineer',
   Date_of_Joining: ISODate('2019-06-15T00:00:00.000Z'),
   Salary: 60000,
   Dept_Name: 'Data Science',
   Projects: [ 'AI-Analytics', 'Cloud-Migration', 'BigData-Pipeline' ]
 },
 {
   _id: ObjectId('6989a2be9a3a13ba051e2622'),
   Emp_Id: 122,
   Emp_Name: 'Ravi',
   Designation: 'Senior Engineer',
   Date_of_Joining: ISODate('2018-03-20T00:00:00.000Z'),
   Salary: 75000,
   Dept_Name: 'R&D'
 },
 {
   _id: ObjectId('6989a2be9a3a13ba051e2623'),
   Emp_Id: 123,
   Emp_Name: 'Meera',
   Designation: 'Manager',
   Date_of_Joining: ISODate('2016-01-10T00:00:00.000Z'),
   Salary: 90000,
   Dept_Name: 'HR'
 }

mployee> db.Employee_Info.insertOne({
..   Emp_Id:130,
..   Emp_Name:"TemporaryUser",
..   createdAt:new Date()
.. })
..
```

```
    Emp_Id: 122,
    Emp_Name: 'Ravi',
    Designation: 'Senior Engineer',
    Date_of_Joining: ISODate('2018-03-20T00:00:00.000Z'),
    Salary: 75000,
    Dept_Name: 'R&D'
  },
  {
    _id: ObjectId('6989a2be9a3a13ba051e2623'),
    Emp_Id: 123,
    Emp_Name: 'Meera',
    Designation: 'Manager',
    Date_of_Joining: ISODate('2016-01-10T00:00:00.000Z'),
    Salary: 90000,
    Dept_Name: 'HR'
  }
]
Employee> db.Employee_Info.insertOne({
...   Emp_Id:130,
...   Emp_Name:"TemporaryUser",
...   createdAt:new Date()
... })
...
{
  acknowledged: true,
  insertedId: ObjectId('6989a3909a3a13ba051e2624')
}
Employee> db.Employee_Info.createIndex(
...   {createdAt:1},
...   {expireAfterSeconds:15}
... )
...
createdAt_1
Employee> db.Employee_Info.find({Emp_Id:130})
[
  {
    _id: ObjectId('6989a3909a3a13ba051e2624'),
    Emp_Id: 130,
    Emp_Name: 'TemporaryUser',
    createdAt: ISODate('2026-02-09T09:06:24.471Z')
  }
]
Employee> show dbs
Employee       108.00 KiB
Employee_Info   72.00 KiB
admin           40.00 KiB
config         108.00 KiB
e_commerce_db  120.00 KiB
local           96.00 KiB
travelmate      40.00 KiB
Employee>
```

# LABORATORY PROGRAM – 3

## Perform the following DB operations using Cassandra

### Questions:

a) Create a keyspace by name Library
b) Create a column family by name Library-Info with attributes

- Stud_Id Primary Key,
- Counter_value of type Counter,
- Stud_Name, Book-Name, Book-Id,
- Date_of_issue

c) Insert the values into the table in batch
d) Display the details of the table created and increase the value of the counter
e) Write a query to show that a student with id 112 has taken a book "BDA" 2 times.
f) Export the created column to a csv file
g) Import a given csv dataset from local file system into Cassandra column family

COMMAND WITH OUTPUT

```
cqlsh> CREATE KEYSPACE Library
   ... WITH replication = {
   ...     'class': 'SimpleStrategy',
   ...     'replication_factor': 1
   ... };
```

```
cqlsh> USE Library;
```

```
cqlsh:library> CREATE TABLE Library_Info (
           ...     Stud_Id int,
           ...     Stud_Name text,
           ...     Book_Name text,
           ...     Book_Id int,
           ...     Date_of_issue date,
           ...     PRIMARY KEY (Stud_Id, Book_Id)
           ... );
cqlsh:library> CREATE TABLE Library_Counter (
           ...     Stud_Id int,
           ...     Book_Name text,
           ...     Counter_value counter,
           ...     PRIMARY KEY (Stud_Id, Book_Name)
           ... );
```

```
cqlsh:library> BEGIN BATCH
           ...
           ... INSERT INTO Library_Info (Stud_Id, Stud_Name, Book_Name, Book_Id,
 Date_of_issue)
           ... VALUES (112, 'Rahul', 'BDA', 101, '2026-02-15');
           ...
           ... INSERT INTO Library_Info (Stud_Id, Stud_Name, Book_Name, Book_Id,
 Date_of_issue)
           ... VALUES (113, 'Amit', 'DBMS', 102, '2026-02-14');
           ...
           ... APPLY BATCH;
```

```
cqlsh:library> UPDATE Library_Counter
            ... SET Counter_value = Counter_value + 1
            ... WHERE Stud_Id = 112 AND Book_Name = 'BDA';
cqlsh:library> UPDATE Library_Counter
            ... SET Counter_value = Counter_value + 1
            ... WHERE Stud_Id = 112 AND Book_Name = 'BDA';
```

```
cqlsh:library> SELECT * FROM Library_Info;

 stud_id | book_id | book_name | date_of_issue | stud_name
---------+---------+-----------+---------------+-----------
     113 |     102 |      DBMS |    2026-02-14 |      Amit
     112 |     101 |       BDA |    2026-02-15 |     Rahul

(2 rows)
cqlsh:library> SELECT * FROM Library_Counter;

 stud_id | book_name | counter_value
---------+-----------+---------------
     112 |       BDA |             2
```

```
cqlsh:library> SELECT Counter_value
            ... FROM Library_Counter
            ... WHERE Stud_Id = 112 AND Book_Name = 'BDA';

 counter_value
---------------
             2
```

```
Starting copy of library.library_info with columns [stud_id, book_id, book_name,
 date_of_issue, stud_name].
Processed: 2 rows; Rate:      14 rows/s; Avg. rate:      14 rows/s
2 rows exported to 1 files in 0.169 seconds.
cqlsh:library> COPY Library_Counter TO 'library_counter.csv' WITH HEADER = TRUE;
Using 16 child processes

Starting copy of library.library_counter with columns [stud_id, book_name, count
er_value].
Processed: 1 rows; Rate:      10 rows/s; Avg. rate:      10 rows/s
1 rows exported to 1 files in 0.120 seconds.
cqlsh:library>
```

```
cqlsh:library> COPY Library_Info
           ... (stud_id, book_id, book_name, date_of_issue, stud_name)
           ... FROM 'library_info.csv'
           ... WITH HEADER = TRUE;
Using 16 child processes

Starting copy of library.library_info with columns [stud_id, book_id, book_name,
 date_of_issue, stud_name].
Processed: 2 rows; Rate:       4 rows/s; Avg. rate:       5 rows/s
2 rows imported from 1 files in 0.387 seconds (0 skipped).
cqlsh:library> SELECT * FROM Library_Info;

 stud_id | book_id | book_name | date_of_issue | stud_name
---------+---------+-----------+---------------+-----------
     113 |     102 |      DBMS |    2026-02-14 |      Amit
     112 |     101 |       BDA |    2026-02-15 |     Rahul

(2 rows)
```

# LABORATORY PROGRAM – 4

## Execution of HDFS Commands for interaction with Hadoop Environment. (Minimum 10 commands to be executed)

COMMAND WITH OUTPUT

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ cd ./Desktop/
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ start-all.sh
WARNING: Attempting to start all Apache Hadoop daemons as hadoop in 10 seconds.
WARNING: This is not a recommended production deployment configuration.
WARNING: Use CTRL-C to abort.
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [bmscecse-HP-Elite-Tower-800-G9-Desktop-PC]
Starting resourcemanager
Starting nodemanagers
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hdfs dfs -mkdir /Lab05
```

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hadoop fs -ls /Hadoop
ls: `/Hadoop': No such file or directory
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hadoop fs -ls /Lab05
```

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ touch test.txt
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ nano text.txt
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hdfs dfs -put ./text.txt /Lab05/text.txt
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hadoop fs -ls /Lab05
Found 1 items
-rw-r--r--   1 hadoop supergroup         19 2024-05-13 14:33 /Lab05/text.txt
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hdfs dfs -cat /Lab05/text.txt
Hello
How are you?
```

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hadoop fs -ls /Lab05
Found 2 items
-rw-r--r--   1 hadoop supergroup         15 2024-05-13 14:40 /Lab05/test.txt
-rw-r--r--   1 hadoop supergroup         19 2024-05-13 14:33 /Lab05/text.txt
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hdfs dfs -getmerge /Lab05 /text.txt /Lab05 /test.txt ../
Downloads/Merged.txt
getmerge: `/text.txt': No such file or directory
getmerge: `/test.txt': No such file or directory
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hdfs dfs -getmerge /Lab05/text.txt /Lab05/test.txt ../Do
wnloads/Merged.txt
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hadoop fs -getfacl /Lab05
# file: /Lab05
# owner: hadoop
# group: supergroup
user::rwx
group::r-x
other::r-x
```

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hdfs dfs -put /home/hadoop/Desktop/Welcome.txt /abc/WC.txt
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hdfs dfs -copyFromLocal /home/hadoop/Desktop/Welcome.txt /abc/WC.txt
copyFromLocal: `/abc/WC.txt': File exists
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hdfs dfs -get /abc/WC.txt /home/hadoop/Downloads/WWC.txt
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hdfs dfs -getmerge /abc/ /home/hadoop/Desktop/Merge.txt
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hadoop fs -getfacl /abc/
# file: /abc
# owner: hadoop
# group: supergroup
user::rwx
group::r-x
other::r-x

hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hdfs dfs -cat /abc/WC.txt
hello world
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hadoop fs -mv /abc /FFF
hadoop fs -ls /FFF
Found 3 items
-rw-r--r--   1 hadoop supergroup         12 2025-04-15 14:53 /FFF/WC.txt
-rw-r--r--   1 hadoop supergroup         12 2024-05-14 14:35 /FFF/file.txt
-rw-r--r--   1 hadoop supergroup         12 2024-05-14 14:38 /FFF/file_cp_local.txt
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hadoop fs -cp /CSE/ /LLL
```

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hdfs dfs -copyToLocal /Lab05/text.txt ../Documents
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hdfs dfs -copyToLocal /Lab05/test.txt ../Documents
```

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hdfs dfs -cat /Lab05/text.txt
Hello
How are you?
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hdfs dfs -mv /Lab05 /test_Lab05
```

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hdfs dfs -ls /test_Lab05
Found 2 items
-rw-r--r--   1 hadoop supergroup         15 2024-05-13 14:40 /test_Lab05/test.txt
-rw-r--r--   1 hadoop supergroup         19 2024-05-13 14:33 /test_Lab05/text.txt
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hdfs dfs -cp /test_Lab05/ /Lab05
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hdfs dfs -ls /Lab05
Found 2 items
-rw-r--r--   1 hadoop supergroup         15 2024-05-13 14:51 /Lab05/test.txt
-rw-r--r--   1 hadoop supergroup         19 2024-05-13 14:51 /Lab05/text.txt
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hdfs dfs -ls /test_Lab05
Found 2 items
-rw-r--r--   1 hadoop supergroup         15 2024-05-13 14:40 /test_Lab05/test.txt
-rw-r--r--   1 hadoop supergroup         19 2024-05-13 14:33 /test_Lab05/text.txt
```

# LABORATORY PROGRAM – 5

## Implement Wordcount program on Hadoop framework

CODE, COMMAND WITH OUTPUT

### Driver Code

```
// Importing libraries
import java.io.IOException;



import org.apache.hadoop.conf.Configured;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapred.FileInputFormat;
import org.apache.hadoop.mapred.FileOutputFormat;
import org.apache.hadoop.mapred.JobClient;
import org.apache.hadoop.mapred.JobConf;
import org.apache.hadoop.util.Tool;
import org.apache.hadoop.util.ToolRunner;

public class WCDriver extends Configured implements Tool {

    public int run(String[] args) throws IOException {
        if (args.length < 2) {
            System.out.println("Please give valid inputs");
            return -1;
        }

        JobConf conf = new JobConf(WCDriver.class);
        conf.setJobName("WordCount");

        FileInputFormat.setInputPaths(conf, new Path(args[0]));
        FileOutputFormat.setOutputPath(conf, new Path(args[1]));

        conf.setMapperClass(WCMapper.class);
        conf.setReducerClass(WCReducer.class);

        conf.setMapOutputKeyClass(Text.class);
        conf.setMapOutputValueClass(IntWritable.class);

        conf.setOutputKeyClass(Text.class);
        conf.setOutputValueClass(IntWritable.class);

        JobClient.runJob(conf);
        return 0;
    }

    // Main Method
    public static void main(String[] args) throws Exception {
        int exitCode = ToolRunner.run(new WCDriver(), args);
        System.out.println("Job Exit Code: " + exitCode);
    }
}
```

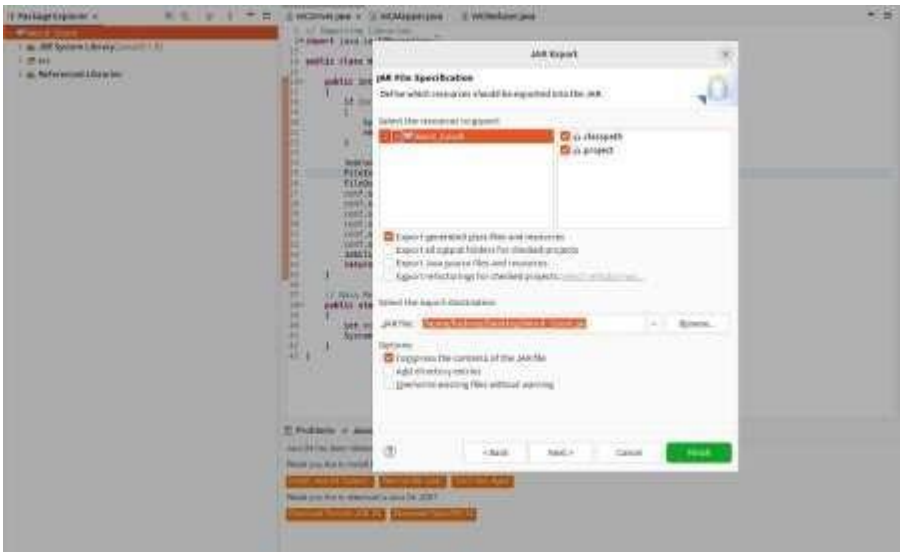
### Mapper Code

```
// Importing libraries
import java.io.IOException;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapred.MapReduceBase;
import org.apache.hadoop.mapred.Mapper;
import org.apache.hadoop.mapred.OutputCollector;
import org.apache.hadoop.mapred.Reporter;
```

```
public class WCMapper extends MapReduceBase implements Mapper<LongWritable, Text, Text, IntWritable> {

    // Map function
    public void map(LongWritable key, Text value, OutputCollector<Text, IntWritable> output, Reporter reporter)
            throws IOException {
        String line = value.toString();

        // Splitting the line on whitespace
        for (String word : line.split("\\s+")) {
            if (word.length() > 0) {
                output.collect(new Text(word), new IntWritable(1));
            }
        }
    }
}
```

## Reducer Code

```
// Importing libraries
import java.io.IOException;
import java.util.Iterator;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapred.MapReduceBase;
import  org.apache.hadoop.mapred.OutputCollector;
import org.apache.hadoop.mapred.Reducer;
import org.apache.hadoop.mapred.Reporter;

public class WCReducer extends MapReduceBase implements Reducer<Text, IntWritable, Text, IntWritable> {

    // Reduce function
    public void reduce(Text key, Iterator<IntWritable> values,
                    OutputCollector<Text, IntWritable> output,
                    Reporter reporter) throws IOException {
        int count = 0;

        // Counting the frequency of each word
        while (values.hasNext()) {
            count += values.next().get();
        }

        output.collect(key, new IntWritable(count));
    }
}
```



```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ cd ./Desktop/
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ start-all.sh
WARNING: Attempting to start all Apache Hadoop daemons as hadoop in 10 seconds.
WARNING: This is not a recommended production deployment configuration.
WARNING: Use CTRL-C to abort.
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [bmscecse-HP-Elite-Tower-800-G9-Desktop-PC]
Starting resourcemanager
Starting nodemanagers
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hdfs dfs -mkdir /Lab06
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hadoop fs -ls /Lab06
```

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ jps
7360 DataNode
7928 ResourceManager
8681 Jps
7178 NameNode
8091 NodeManager
7644 SecondaryNameNode
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ cd ..
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ cd ./Desktop/
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ nano file1.txt
```

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hadoop fs -copyFromLocal -f /home/hadoop/Desktop/file1.txt /rgs/test.txt
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hadoop jar /home/hadoop/Desktop/WordCount.jar wordcount.WordCount /rgs/test.txt /output
JAR does not exist or is not a normal file: /home/hadoop/Desktop/WordCount.jar
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hadoop jar /home/hadoop/Desktop/Word_Count.jar wordcount.WordCount /rgs/test.txt /output
Exception in thread "main" java.lang.ClassNotFoundException: wordcount.WordCount
        at java.base/java.net.URLClassLoader.findClass(URLClassLoader.java:476)
        at java.base/java.lang.ClassLoader.loadClass(ClassLoader.java:594)
        at java.base/java.lang.ClassLoader.loadClass(ClassLoader.java:527)
        at java.base/java.lang.Class.forName0(Native Method)
        at java.base/java.lang.Class.forName(Class.java:398)
        at org.apache.hadoop.util.RunJar.run(RunJar.java:321)
        at org.apache.hadoop.util.RunJar.main(RunJar.java:241)
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hadoop fs -cat /output/part-00000
are     1
brother 1
family  1
hi      1
how     5
is      4
job     1
sister  1
you     1
your    4
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hadoop fs -ls /output
Found 2 items
-rw-r--r--   1 hadoop supergroup          0 2024-05-21 15:21 /output/_SUCCESS
-rw-r--r--   1 hadoop supergroup         69 2024-05-21 15:21 /output/part-00000
```

# LABORATORY PROGRAM – 6

## Implement Weather program on Hadoop framework

## Questions:

From the following link extract the weather data
https://github.com/tomwhite/hadoopbook/tree/master/input/ncdc/all
a) Create a MapReduce program to find average temperature for each year from NCDC data set.
b) find the mean max temperature for every month.

CODE, COMMAND WITH OUTPUT – A

**Driver Code**

```
package temp;



import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;

import org.apache.hadoop.mapreduce.Job;

import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class AverageDriver {

    public static void main(String[] args) throws Exception {

        if (args.length != 2) {
            System.err.println("Please enter both input and output parameters.");
            System.exit(-1);
        }

        // Creating a configuration and job instance
        Configuration conf = new Configuration();
        Job job = Job.getInstance(conf, "Average Calculation");

        job.setJarByClass(AverageDriver.class);

        // Input and output paths
        FileInputFormat.addInputPath(job, new Path(args[0]));
        FileOutputFormat.setOutputPath(job, new Path(args[1]));

        // Setting mapper and reducer classes
        job.setMapperClass(AverageMapper.class);
        job.setReducerClass(AverageReducer.class);

        // Output key and value types
        job.setOutputKeyClass(Text.class);
        job.setOutputValueClass(IntWritable.class);

        // Submitting the job and waiting for it to complete
        System.exit(job.waitForCompletion(true) ? 0 : 1);
    }
}
```

## Mapper Code

```
package temp;

import java.io.IOException;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Mapper;

public class AverageMapper extends Mapper<LongWritable, Text, Text, IntWritable> {

    public static final int MISSING = 9999;

    @Override
    public void map(LongWritable key, Text value, Context context)
            throws IOException, InterruptedException {

        String line = value.toString();

        // Extract year from fixed position
        String year = line.substring(15, 19);
        int temperature;

        // Determine if there's a '+' sign
        if (line.charAt(87) == '+') {
            temperature = Integer.parseInt(line.substring(88, 92));
        } else {
            temperature = Integer.parseInt(line.substring(87, 92));
        }

        // Quality check character
        String quality = line.substring(92, 93);

        // Only emit if data is valid
        if (temperature != MISSING && quality.matches("[01459]")) {
            context.write(new Text(year), new IntWritable(temperature));
        }
    }
}
```

## Reducer Code

```
package temp;

import java.io.IOException;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;

public class AverageReducer extends Reducer<Text, IntWritable, Text, IntWritable> {

    @Override
    public void reduce(Text key, Iterable<IntWritable> values,
                    Context context) throws IOException, InterruptedException {

        int sumTemp = 0;
        int count = 0;

        for (IntWritable value : values) {
            sumTemp += value.get();
            count++;
        }

        if (count > 0) {
            int average = sumTemp / count;
            context.write(key, new IntWritable(average));
```

```
        }
    }
}
```

| Name | Size | Type | Modified |
|---|---|---|---|
| META-INF | 25 bytes | Folder | |
| .classpath | 2.2 kB | unknown | 06 May 2025, 14:40 |
| .project | 377 bytes | unknown | 06 May 2025, 14:34 |
| AverageDriver.class | 1.6 kB | Java class | 06 May 2025, 14:42 |
| AverageMapper.class | 2.4 kB | Java class | 06 May 2025, 14:42 |
| AverageReducer.class | 2.3 kB | Java class | 06 May 2025, 14:42 |

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ start-all.sh
WARNING: Attempting to start all Apache Hadoop daemons as hadoop in 10 seconds.
WARNING: This is not a recommended production deployment configuration.
WARNING: Use CTRL-C to abort.
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [bmscecse-HP-Elite-Tower-800-G9-Desktop-PC]
Starting resourcemanager
Starting nodemanagers
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ jps
7056 DataNode
7332 SecondaryNameNode
7638 ResourceManager
8231 Jps
5883 org.eclipse.equinox.launcher_1.6.1000.v20250227-1734.jar
7884 NodeManager
6877 NameNode
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -ls /\
> ^C
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -ls /
Found 4 items
drwxr-xr-x   - hadoop supergroup          0 2025-04-15 15:00 /FFF
drwxr-xr-x   - hadoop supergroup          0 2025-04-15 15:34 /LLL
drwxr-xr-x   - hadoop supergroup          0 2024-05-13 14:46 /file
drwxr-xr-x   - hadoop supergroup          0 2024-05-13 15:18 /newDataFlair
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -ls /weather
ls: `/weather': No such file or directory
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -mkdir /weather
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -copyFromLocal /home/hadoop/Desktop/1901.txt /weather/test.txt
```

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop jar /home/hadoop/Desktop/AverageTemperature.jar AverageDriver /weather/test.txt /weather/output
2025-05-06 14:59:23,230 INFO impl.MetricsConfig: Loaded properties from hadoop-metrics2.properties
2025-05-06 14:59:23,279 INFO impl.MetricsSystemImpl: Scheduled Metric snapshot period at 10 second(s).
2025-05-06 14:59:23,279 INFO impl.MetricsSystemImpl: JobTracker metrics system started
2025-05-06 14:59:23,348 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2025-05-06 14:59:23,399 INFO input.FileInputFormat: Total input files to process : 1
2025-05-06 14:59:23,422 INFO mapreduce.JobSubmitter: number of splits:1
2025-05-06 14:59:23,467 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_local1822813_0001
2025-05-06 14:59:23,468 INFO mapreduce.JobSubmitter: Executing with tokens: []
2025-05-06 14:59:23,568 INFO mapreduce.Job: The url to track the job: http://localhost:8080/
2025-05-06 14:59:23,568 INFO mapreduce.Job: Running job: job_local1822813_0001
2025-05-06 14:59:23,581 INFO mapred.LocalJobRunner: OutputCommitter set in config null
2025-05-06 14:59:23,588 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileOutputCommitterFactory
2025-05-06 14:59:23,595 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2025-05-06 14:59:23,595 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup failures: false
2025-05-06 14:59:23,595 INFO mapred.LocalJobRunner: OutputCommitter is org.apache.hadoop.mapreduce.lib.output.FileOutputCommitter
2025-05-06 14:59:23,601 INFO mapred.LocalJobRunner: Waiting for map tasks
2025-05-06 14:59:23,603 INFO mapred.LocalJobRunner: Starting task: attempt_local1822813_0001_m_000000_0
2025-05-06 14:59:23,615 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileOutputCommitterFactory
2025-05-06 14:59:23,616 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2025-05-06 14:59:23,616 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup failures: false
2025-05-06 14:59:23,622 INFO mapred.Task:  Using ResourceCalculatorProcessTree : [ ]
2025-05-06 14:59:23,624 INFO mapred.MapTask: Processing split: hdfs://localhost:9000/weather/test.txt:0+888100
2025-05-06 14:59:23,658 INFO mapred.MapTask: (EQUATOR) 0 kvi 26214396(104857584)
2025-05-06 14:59:23,658 INFO mapred.MapTask: mapreduce.task.io.sort.mb: 100
2025-05-06 14:59:23,658 INFO mapred.MapTask: soft limit at 83886080
2025-05-06 14:59:23,658 INFO mapred.MapTask: bufstart = 0; bufvoid = 104857600
2025-05-06 14:59:23,658 INFO mapred.MapTask: kvstart = 26214396; length = 6553600
2025-05-06 14:59:23,660 INFO mapred.MapTask: Map output collector class = org.apache.hadoop.mapred.MapTask$MapOutputBuffer
```

```
2025-05-06 14:59:24,581 INFO mapreduce.Job: Counters: 36
        File System Counters
                FILE: Number of bytes read=153118
                FILE: Number of bytes written=1493804
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=1776380
                HDFS: Number of bytes written=8
                HDFS: Number of read operations=15
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=4
                HDFS: Number of bytes read erasure-coded=0
        Map-Reduce Framework
                Map input records=6565
                Map output records=6564
                Map output bytes=59076
                Map output materialized bytes=72210
                Input split bytes=103
                Combine input records=0
                Combine output records=0
                Reduce input groups=1
                Reduce shuffle bytes=72210
                Reduce input records=6564
                Reduce output records=1
                Spilled Records=13128
                Shuffled Maps =1
                Failed Shuffles=0
                Merged Map outputs=1
                GC time elapsed (ms)=0
                Total committed heap usage (bytes)=1266679808
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Input Format Counters
                Bytes Read=888190
        File Output Format Counters
                Bytes Written=8
```

```
                Bytes Written=8
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -ls /weather
Found 2 items
drwxr-xr-x   - hadoop supergroup          0 2025-05-06 14:59 /weather/output
-rw-r--r--   1 hadoop supergroup     888190 2025-05-06 14:50 /weather/test.txt
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -ls /weather/output
Found 2 items
-rw-r--r--   1 hadoop supergroup          0 2025-05-06 14:59 /weather/output/_SUCCESS
-rw-r--r--   1 hadoop supergroup          8 2025-05-06 14:59 /weather/output/part-r-00000
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -cat /weather/output/part-r-00000
1901    46
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$
```

CODE, COMMAND WITH OUTPUT – B

**Driver Code**

```
package meanmax;

import org.apache.hadoop.conf.Configuration; import org.apache.hadoop.fs.Path;

import org.apache.hadoop.io.IntWritable; import org.apache.hadoop.io.Text;


import org.apache.hadoop.mapreduce.Job;


import org.apache.hadoop.mapreduce.lib.input.FileInputFormat; import
org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class MeanMaxDriver {

public static void main(String[] args) throws Exception { if (args.length != 2) {

System.err.println("Please enter both input and output parameters."); System.exit(-
1);

}

Configuration conf = new Configuration();

Job job = Job.getInstance(conf, "Mean and Max Temperature");
job.setJarByClass(MeanMaxDriver.class);

FileInputFormat.addInputPath(job, new Path(args[0]));
FileOutputFormat.setOutputPath(job, new Path(args[1]));

job.setMapperClass(MeanMaxMapper.class);
job.setReducerClass(MeanMaxReducer.class);


job.setOutputKeyClass(Text.class);  job.setOutputValueClass(IntWritable.class);


System.exit(job.waitForCompletion(true) ? 0 : 1);

}

}
```

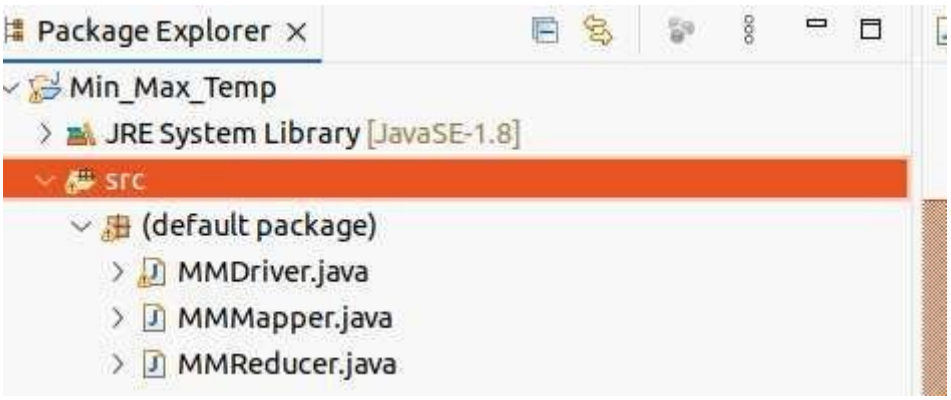
**Mapper Code**

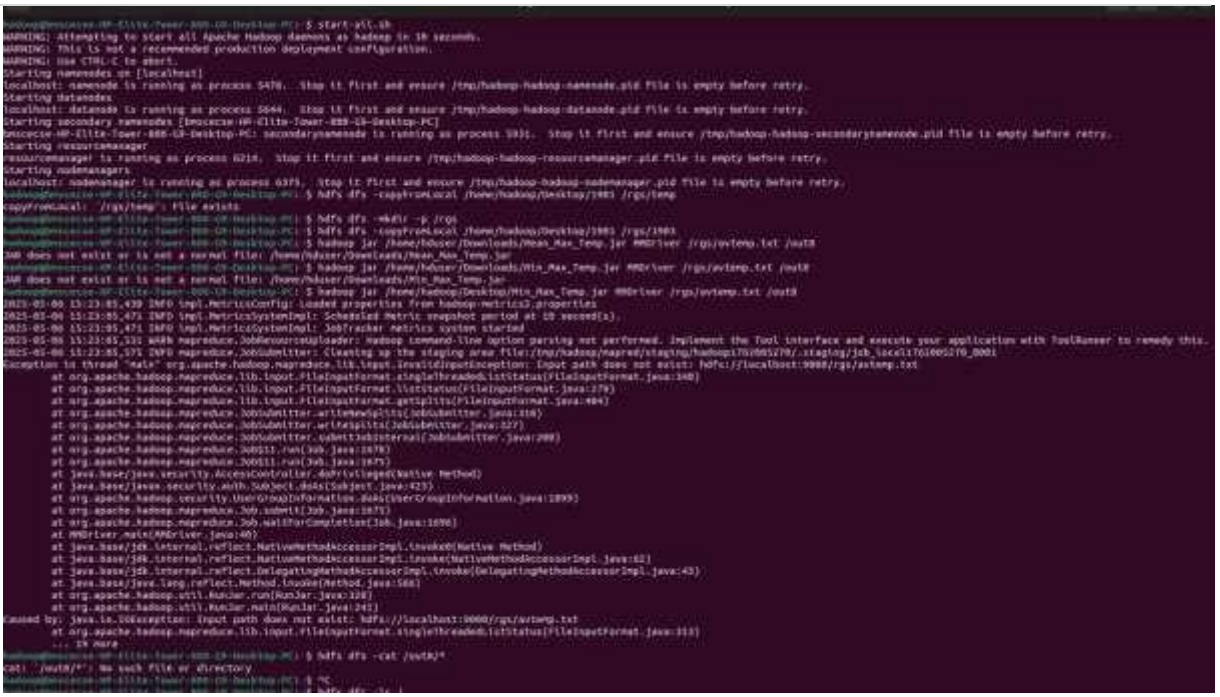
```
package meanmax;

import java.io.IOException;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.LongWritable;
```

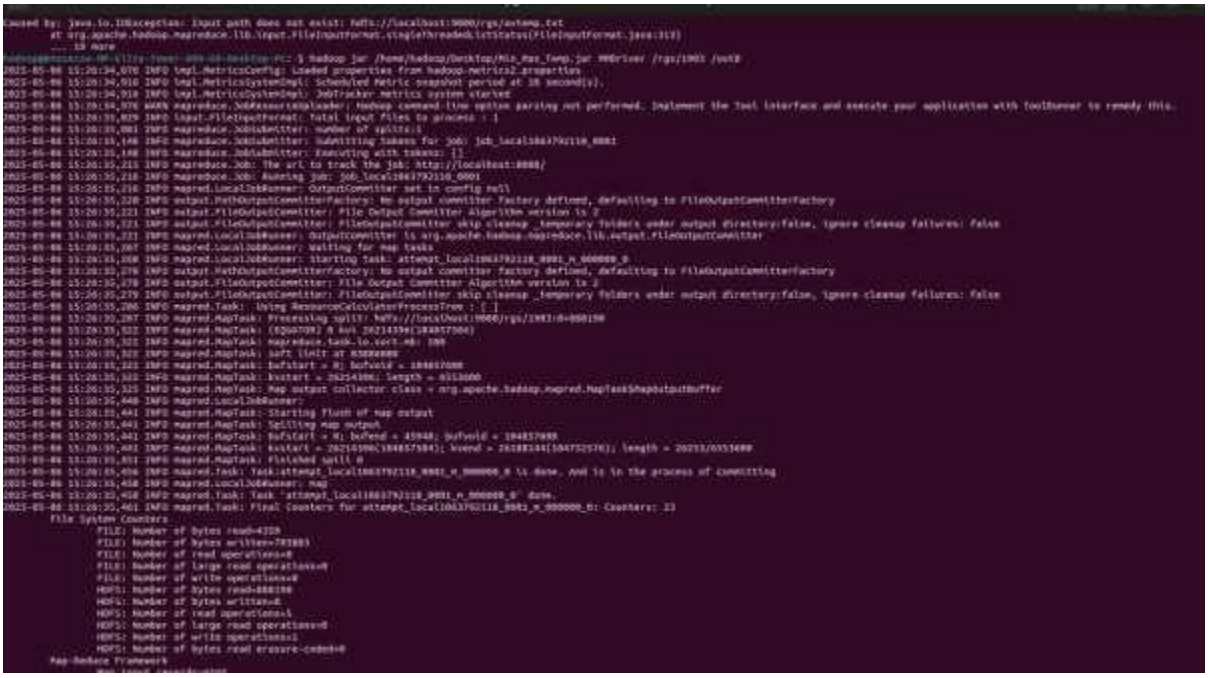
```
import org.apache.hadoop.io.Text;

import org.apache.hadoop.mapreduce.Mapper;

public class MeanMaxMapper extends Mapper<LongWritable, Text, Text, IntWritable> {

    public static final int MISSING = 9999;

    @Override
    public void map(LongWritable key, Text value, Context context)
            throws IOException, InterruptedException {

        String line = value.toString();

        // Extract month from positions 19-20
        String month = line.substring(19, 21);
        int temperature;
        // Extract temperature considering optional '+'
        if (line.charAt(87) == '+') {
            temperature = Integer.parseInt(line.substring(88, 92));
        } else {
            temperature = Integer.parseInt(line.substring(87, 92));
        }

        // Quality check
        String quality = line.substring(92, 93);

        if (temperature != MISSING && quality.matches("[01459]")) {
            context.write(new Text(month), new IntWritable(temperature));
        }
    }
}
```

## Reducer Code

```
package meanmax;

import java.io.IOException;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;

public class MeanMaxReducer extends Reducer<Text, IntWritable, Text, Text> {

    @Override
    public void reduce(Text key, Iterable<IntWritable> values,
                    Context context) throws IOException, InterruptedException {

        int sumTemp = 0;
        int count = 0;
        int maxTemp = Integer.MIN_VALUE;

        for (IntWritable value : values) {
            int temp = value.get();
            sumTemp += temp;
            count++;

            if (temp > maxTemp) {
                maxTemp = temp;
            }
        }

        if (count > 0) {
            int avgTemp = sumTemp / count;
            String result = "mean=" + avgTemp + " max=" + maxTemp;
            context.write(key, new Text(result));
        }
    }
}
```


Package Explorer
Min_Max_Temp
JRE System Library [JavaSE-1.8]
src
(default package)
MMDriver.java
MMMapper.java
MMReducer.java





```
2025-05-06 15:26:36,233 INFO mapreduce.Job: Counters: 36
        File System Counters
                FILE: Number of bytes read=126914
                FILE: Number of bytes written=1466688
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=177638B
                HDFS: Number of bytes written=74
                HDFS: Number of read operations=15
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=4
                HDFS: Number of bytes read erasure-coded=0
        Map-Reduce Framework
                Map input records=6565
                Map output records=6564
                Map output bytes=45948
                Map output materialized bytes=59082
                Input split bytes=95
                Combine input records=0
                Combine output records=0
                Reduce input groups=12
                Reduce shuffle bytes=59082
                Reduce input records=6564
                Reduce output records=12
                Spilled Records=13128
                Shuffled Maps =1
                Failed Shuffles=0
                Merged Map outputs=1
                GC time elapsed (ms)=0
                Total committed heap usage (bytes)=1052770304
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Input Format Counters
                Bytes Read=888190
        File Output Format Counters
                Bytes Written=74
```

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hdfs dfs -cat /out8/*
01      4
02      0
03      7
04      44
05      100
06      168
07      219
08      198
09      141
10      100
11      19
12      3
```

# LABORATORY PROGRAM – 7

## For a given Text file, Create a Map Reduce program to sort the content in an alphabetic order listing only top 10 maximum occurrences of words.

CODE, COMMAND WITH OUTPUT

### Driver Code (TopNDriver.java)

```
package samples.topn;



import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;

import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class TopNDriver {

    public static void main(String[] args) throws Exception {
        if (args.length != 3) {
            System.err.println("Usage: TopNDriver <in> <temp-out> <final-out>");
            System.exit(2);
        }

        Configuration conf = new Configuration();

        // === Job 1: Word Count ===
        Job wcJob = Job.getInstance(conf, "word count");
        wcJob.setJarByClass(TopNDriver.class);
        wcJob.setMapperClass(WordCountMapper.class);
        wcJob.setCombinerClass(WordCountReducer.class);
        wcJob.setReducerClass(WordCountReducer.class);
        wcJob.setOutputKeyClass(Text.class);
        wcJob.setOutputValueClass(IntWritable.class);

        FileInputFormat.addInputPath(wcJob, new Path(args[0]));
        Path tempDir = new Path(args[1]);
        FileOutputFormat.setOutputPath(wcJob, tempDir);

        if (!wcJob.waitForCompletion(true)) {
            System.exit(1);
        }

        // === Job 2: Top N ===
        Job topJob = Job.getInstance(conf, "top 10 words");
        topJob.setJarByClass(TopNDriver.class);
        topJob.setMapperClass(TopNMapper.class);
        topJob.setReducerClass(TopNReducer.class);
        topJob.setMapOutputKeyClass(IntWritable.class);
        topJob.setMapOutputValueClass(Text.class);
        topJob.setOutputKeyClass(Text.class);
        topJob.setOutputValueClass(IntWritable.class);

        FileInputFormat.addInputPath(topJob, tempDir);
        FileOutputFormat.setOutputPath(topJob, new Path(args[2]));

        System.exit(topJob.waitForCompletion(true) ? 0 : 1); }
```

```
}
```

## Mapper Code (WordCountMapper.java)

```
package samples.topn;

import java.io.IOException;
import java.util.StringTokenizer;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Mapper;

public class WordCountMapper
    extends Mapper<Object, Text, Text, IntWritable> {

    private final static IntWritable ONE = new IntWritable(1);
    private Text word = new Text();
    // characters to normalize into spaces
    private String tokens = "[_|$#<>\\^=\\[\\]\\*/\\\\,;,.\\-:()?!\"']";

    @Override
    protected void map(Object key, Text value, Context context)
        throws IOException, InterruptedException {

        // clean & tokenize
        String clean = value.toString()
                        .toLowerCase()
                        .replaceAll(tokens, " ");
        StringTokenizer itr = new StringTokenizer(clean);
        while (itr.hasMoreTokens()) {
            word.set(itr.nextToken().trim());
            context.write(word, ONE);
        }
    }
}
```

## Mapper Code (TopNMapper.java)

```
package samples.topn;

import java.io.IOException;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Mapper;

public class TopNMapper
    extends Mapper<Object, Text, IntWritable, Text> {

    private IntWritable count = new IntWritable();
    private Text word = new Text();

    @Override
    protected void map(Object key, Text value, Context context)
        throws IOException, InterruptedException {

        // input line: word \t count
        String[] parts = value.toString().split("\\t");
        if (parts.length == 2) {
            word.set(parts[0]);
            count.set(Integer.parseInt(parts[1]));
            // emit count → word, so Hadoop sorts by count
            context.write(count, word);
        }
    }
}
```

## Reducer Code (WordCountReducer.java)

```
package samples.topn;

import java.io.IOException;


import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;

public class WordCountReducer
    extends Reducer<Text, IntWritable, Text, IntWritable> {

    @Override
    protected void reduce(Text key, Iterable<IntWritable> values, Context context)
        throws IOException, InterruptedException {

        int sum = 0;
        for (IntWritable val : values) {
            sum += val.get();
        }
        context.write(key, new IntWritable(sum));
    }
}
```

## Reducer Code (TopNReducer.java)

```
package samples.topn;

import java.io.IOException;
import java.util.ArrayList;
import java.util.Collections;
import java.util.List;
import java.util.Map;
import java.util.TreeMap;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;

public class TopNReducer
    extends Reducer<IntWritable, Text, Text, IntWritable> {

    // TreeMap with descending order of keys (counts)
    private TreeMap<Integer, List<String>> countMap =
        new TreeMap<>(Collections.reverseOrder());

    @Override
    protected void reduce(IntWritable key, Iterable<Text> values, Context context)
        throws IOException, InterruptedException {

        int cnt = key.get();
        List<String> words = countMap.getOrDefault(cnt, new ArrayList<>());
        for (Text w : values) {
            words.add(w.toString());
        }
        countMap.put(cnt, words);
    }

    @Override
    protected void cleanup(Context context)
        throws IOException, InterruptedException {

        // collect top 10 word→count pairs
        List<WordCount> topList = new ArrayList<>();
        int seen = 0;
        for (Map.Entry<Integer, List<String>> entry : countMap.entrySet()) {
```

```
                int cnt = entry.getKey();
                for (String w : entry.getValue()) {
                    topList.add(new WordCount(w, cnt));
                    seen++;
                    if (seen == 10) break;
                }
if (seen == 10) break;
            }

            // sort these 10 entries alphabetically by word
            Collections.sort(topList, (a, b) -> a.word.compareTo(b.word));

            // emit final top 10 in alphabetical order
            for (WordCount wc : topList) {
                context.write(new Text(wc.word), new IntWritable(wc.count));
            }
        }

        // helper class
        private static class WordCount {
            String word;
            int count;
            WordCount(String w, int c) { word = w; count = c; }
        }
    }
```

```
C:\hadoop-3.3.0\sbin>jps
11072 DataNode
20528 Jps
5620 ResourceManager
15532 NodeManager
6140 NameNode

C:\hadoop-3.3.0\sbin>hdfs dfs -mkdir /input_dir

C:\hadoop-3.3.0\sbin>hdfs dfs -ls /
Found 1 items
drwxr-xr-x   - Anusree supergroup          0 2021-05-08 19:46 /input_dir

C:\hadoop-3.3.0\sbin>hdfs dfs -copyFromLocal C:\input.txt /input_dir

C:\hadoop-3.3.0\sbin>hdfs dfs -ls /input_dir
Found 1 items
-rw-r--r--   1 Anusree supergroup         36 2021-05-08 19:48 /input_dir/input.txt

C:\hadoop-3.3.0\sbin>hdfs dfs -cat /input_dir/input.txt
hello
world
hello
hadoop
bye
```

```
C:\hadoop-3.3.0\sbin>hadoop jar C:\sort.jar samples.topn.TopN /input_dir/input.txt /output_dir
2021-05-08 19:54:54,582 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2021-05-08 19:54:55,291 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/Anusree/.staging/job_1620483374279_0001
2021-05-08 19:54:55,821 INFO input.FileInputFormat: Total input files to process : 1
2021-05-08 19:54:56,261 INFO mapreduce.JobSubmitter: number of splits:1
2021-05-08 19:54:56,552 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1620483374279_0001
2021-05-08 19:54:56,552 INFO mapreduce.JobSubmitter: Executing with tokens: []
2021-05-08 19:54:56,843 INFO conf.Configuration: resource-types.xml not found
2021-05-08 19:54:56,843 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2021-05-08 19:54:57,387 INFO impl.YarnClientImpl: Submitted application application_1620483374279_0001
2021-05-08 19:54:57,507 INFO mapreduce.Job: The url to track the job: http://LAPTOP-JG329ESD:8088/proxy/application_1620483374279_0001/
2021-05-08 19:54:57,508 INFO mapreduce.Job: Running job: job_1620483374279_0001
2021-05-08 19:55:13,792 INFO mapreduce.Job: Job job_1620483374279_0001 running in uber mode : false
2021-05-08 19:55:13,794 INFO mapreduce.Job:  map 0% reduce 0%
2021-05-08 19:55:20,820 INFO mapreduce.Job:  map 100% reduce 0%
2021-05-08 19:55:27,116 INFO mapreduce.Job:  map 100% reduce 100%
2021-05-08 19:55:33,199 INFO mapreduce.Job: Job job_1620483374279_0001 completed successfully
2021-05-08 19:55:33,334 INFO mapreduce.Job: Counters: 54
        File System Counters
                FILE: Number of bytes read=65
                FILE: Number of bytes written=530397
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=142
                HDFS: Number of bytes written=31
                HDFS: Number of read operations=8
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=2
                HDFS: Number of bytes read erasure-coded=0
```

```
C:\hadoop-3.3.0\sbin>hdfs dfs -cat /output_dir/*
hello   2
hadoop  1
world   1
bye     1

C:\hadoop-3.3.0\sbin>
```

# LABORATORY PROGRAM – 8

## Write a Scala program to print numbers from 1 to 100 using for loop.

CODE, COMMAND WITH OUTPUT

```
object PrintNumbers {
  def main(args: Array[String]): Unit = {

    for(i <- 1 to 100) {
      println(i)
    }

  }
}
```

## LABORATORY PROGRAM – 9

**Using RDD and FlatMap count how many times each word appears in a file and write out a list of words whose count is strictly greater than 4 using Spark.**

CODE, COMMAND WITH OUTPUT

```
val textFile = sc.textFile("/home/bhoom/Desktop/wc.txt")

val counts = textFile.flatMap(line => line.split(" ")).map(word => (word,1)).reduceByKey(_
+ _)

import scala.collection.immutable.ListMap

 val sorted=ListMap(counts.collect.sortWith(_._2 > _._2):_*)// sort in descending order
based on values

println(sorted)

for((k,v)<-sorted) {

if(v>4) {

print(k+",")

print(v)

println()

}

}
```

```
scala> val rdd = spark.sparkContext.textFile("file:/home/bmscecse/Desktop/scala")
rdd: org.apache.spark.rdd.RDD[String] = file:/home/bmscecse/Desktop/scala MapPartitionsRDD[1] at textFile at <console>:23

scala> val counts = rdd.flatMap(_.split("\\s+")).map(word => (word.toLowerCase, 1)).reduceByKey(_ + _).filter(_._2 > 4)
counts: org.apache.spark.rdd.RDD[(String, Int)] = MapPartitionsRDD[5] at filter at <console>:25

scala> counts.collect().foreach{ case (word, count) => println(s"$word $count") }
spark 6

scala>
```

# LABORATORY PROGRAM – 10

## Write a simple streaming program in Spark to receive text data streams on a particular port, perform basic text cleaning (like white space removal, stop words removal, lemmatization, etc.), and print the cleaned text on the screen. (Open Ended Question).

CODE, COMMAND WITH OUTPUT

```
# Install NLTK and download required data (run once)
!pip install nltk

import nltk
nltk.download('punkt')
nltk.download('stopwords')
nltk.download('wordnet')

from pyspark.sql import SparkSession
from pyspark.sql.functions import col, lower, regexp_replace, split, explode, udf
from pyspark.sql.types import ArrayType, StringType
from pyspark.ml.feature import StopWordsRemover
from nltk.stem import WordNetLemmatizer

# Initialize SparkSession
spark = SparkSession.builder.appName("TextProcessing").getOrCreate()

# Define your input lines
lines = [
    "Hello, I hate you.",
    "I hate that I love you.",
    "Don't want to, but I can't put",
    "nobody else above you."
]

# Create DataFrame from lines
df = spark.createDataFrame(lines, "string").toDF("value")

# Step 1: Lowercase and remove punctuation
df_clean = df.select(regexp_replace(lower(col("value")), "[^a-zA-Z\\s]", "").alias("cleaned"))

# Step 2: Tokenize the cleaned text
df_tokens = df_clean.select(split(col("cleaned"), "\\s+").alias("tokens"))

# Step 3: Remove stop words
remover = StopWordsRemover(inputCol="tokens", outputCol="filtered")
df_filtered = remover.transform(df_tokens)

# Step 4: Lemmatization using NLTK WordNetLemmatizer with UDF
lemmatizer = WordNetLemmatizer()

def lemmatize_words(words):
    return [lemmatizer.lemmatize(word) for word in words]

lemmatize_udf = udf(lemmatize_words, ArrayType(StringType()))

df_lemmatized = df_filtered.withColumn("lemmatized", lemmatize_udf(col("filtered")))
```

# Step 5: Explode the lemmatized words and show results
df_lemmatized.select(explode(col("lemmatized")).alias("word")).show(truncate=False)

```
Requirement already satisfied: nltk in /usr/local/lib/python3.11/dist-packages (3.9.1)
Requirement already satisfied: click in /usr/local/lib/python3.11/dist-packages (from nltk) (8.2.0)
Requirement already satisfied: joblib in /usr/local/lib/python3.11/dist-packages (from nltk) (1.5.0)
Requirement already satisfied: regex>=2021.8.3 in /usr/local/lib/python3.11/dist-packages (from nltk) (2024.11.6)
Requirement already satisfied: tqdm in /usr/local/lib/python3.11/dist-packages (from nltk) (4.67.1)
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt.zip.
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Unzipping corpora/stopwords.zip.
[nltk_data] Downloading package wordnet to /root/nltk_data...
+------+
|word  |
+------+
|hello |
|hate  |
|hate  |
|love  |
|dont  |
|want  |
|cant  |
|put   |
|nobody|
|else  |
+------+
```